LESSON 01 / 05

The theorem is a census of bracket choices

Expanding $(a+b)^n$ means picking one letter from each of n brackets. A pick-list is determined by which k brackets contributed b: $\binom{n}{k}$ lists, each giving $a^{n-k}b^k$. Hence, for non-negative integer n,

$$(a+b)^n = \sum_{k=0}^{n} \binom{n}{k} a^{n-k}b^k, \qquad T_{k+1} = \binom{n}{k} a^{n-k}b^k$$

No induction, nothing to memorise: the coefficient **is** a count (chapter 8). Symmetry $\binom{n}{k} = \binom{n}{n-k}$ (choose the a-brackets instead) and Pascal's rule (condition on the first bracket) are the same picture. Row n of Pascal's triangle is the census for $(a+b)^n$.

KEY RESULT

The general-term machine

For $(\alpha x^p + \beta x^{-q})^n$ the general term is $\binom{n}{k}\alpha^{n-k}\beta^k x^{p(n-k)-qk}$. "Coefficient of x^m" and "term independent of x" are one step: set the exponent $pn-(p+q)k$ equal to the target, solve for k; integer in $[0,n]$ ⇒ substitute; otherwise **the answer is 0** — "no such term" is a legitimate exam answer.

WORKED EXAMPLE

Worked — the machine at three gears

Coefficient of x^4 in $(2+x)^5$: $k=4$: $\binom{5}{4}2^1 = 10$ (full row $32+80x+80x^2+40x^3+10x^4+x^5$ ✓).

Term independent of x in $\left(x-\frac{2}{x^2}\right)^9$: exponent $9-3k=0$, $k=3$: $\binom{9}{3}(-2)^3 = -672$.

Coefficient of x^4 in $(3-2x)^7$: $k=4$: $\binom{7}{4}3^3(-2)^4 = 35\cdot 27\cdot 16 = 15120$ (even k, positive ✓).

KEY RESULT

Middle terms and equal coefficients

n even: one middle term $T_{n/2+1}$. n odd: two, at $k=\frac{n-1}{2}, \frac{n+1}{2}$. Equal coefficients: $\binom{n}{r} = \binom{n}{s}$ iff $r=s$ or $r+s=n$ — always report both branches: in $(1+x)^{18}$, coefficients of x^{2r} and x^{r+6} equal gives $2r=r+6$ or $2r+r+6=18$, so $r=6$ or $r=4$. Consecutive-coefficient ratios: $\frac{\binom{n}{k+1}}{\binom{n}{k}} = \frac{n-k}{k+1}$ (three consecutive coefficients $36, 84, 126$ force row $n=9$: ratios $\frac{7}{3}, \frac{3}{2}$ ✓).

COMMON TRAP

Common traps — general term

- T_k vs T_{k+1}: the 5th term has $k = 4$.
- Buried negatives: rewrite the bracket as $A + B$ with the minus inside B, then apply the formula robotically.
- Expanding term-by-term instead of writing T_{k+1}: the census view gives each term in one line.

ASSUMPTIONS & CONVENTIONS

Standing conventions for this chapter

Positive integral index n in Part I (official syllabus); general real index is Part II. $\binom{n}{k} = 0$ outside $0 \le k \le n$. Coefficients of an expansion are read after collecting like powers — "number of terms" questions must state in which variable(s) collection happens.

LESSON 02 / 05

Beyond two outcomes per bracket

KEY RESULT

Multinomial coefficients

In $(x_1 + \cdots + x_r)^n$, the coefficient of $x_1^{n_1} \cdots x_r^{n_r}$ $(\sum n_i = n)$ is $\frac{n!}{n_1! \cdots n_r!}$: line up the brackets, mark which gave each outcome. Number of raw products for $(a+b+c)^6$: $3^6 = 729$; collected terms (exponent tuples, stars and bars): $\binom{6+2}{2} = 28$. Coefficient of $a^2b^2c^2$ in $(a+b+c)^6$: $\frac{6!}{2!\,2!\,2!} = 90$.

Coefficient of x^3 in $(1+x+x^2)^4$ by census: patterns $\{x^2, x, 1, 1\}$ in $\frac{4!}{1!\,1!\,2!} = 12$ ways and $\{x, x, x, 1\}$ in $\frac{4!}{3!\,1!} = 4$ ways: total 16 (matches $(1+x+x^2)^4 = 1 + 4x + 10x^2 + 16x^3 + 19x^4 + \cdots$ ✓).

COMMON TRAP

Number-of-terms trap

"Terms in $(1+x+x^2)^n$" is **not** $\binom{n+2}{2}$: collecting in the single variable x collapses to powers $x^0, \ldots, x^{2n}$ — exactly $2n+1$ terms. Always ask: collected in which variables?

KEY RESULT

Where coefficients peak

$\frac{\binom{n}{k+1}}{\binom{n}{k}} = \frac{n-k}{k+1}$ passes 1 at $k = \frac{n-1}{2}$. So row n: n even ⇒ single peak $\binom{n}{n/2}$; n odd ⇒ twin peaks $\binom{n}{(n-1)/2} = \binom{n}{(n+1)/2}$ (for $(1+x)^{13}$: 1716 at x^6, x^7; for $(1+x)^{21}$: 352716 at $k = 10, 11$). Coefficient of x^5 in $(1+x)^{12}$ is $\binom{12}{5} = 792$, and of x^7 also 792 by symmetry.

KEY RESULT

Substitution engine — coefficient sums without expanding

For $f(x) = (px+q)^n$ expanded: sum of coefficients $= f(1) = (p+q)^n$; constant term $= f(0) = q^n$; alternating sum $= f(-1)$ splits even/odd powers. Sum of coefficients of $(2x-1)^7$ is $(2-1)^7 = 1$; of $(x-2y)^9$ (set $x = y = 1$): $(-1)^9 = -1$ — negative answers are legal. Odd-power coefficients of $(1+x)^{12}$: $\frac{f(1)-f(-1)}{2} = 2^{11} = 2048$.

WORKED EXAMPLE

Worked — coefficient extraction across factors

Coefficient of x^8 in $(1+x)^{10}\left(1+\frac{1}{x}\right)^{12}$: write the second factor as $x^{-12}(1+x)^{12}$; the product is $x^{-12}(1+x)^{22}$; need x^{20}: $\binom{22}{20} = 231$. Always clear negative powers into an x^{-m} prefactor first.

LESSON 03 / 05

One master polynomial, three engines

Everything runs on $f(x)=(1+x)^n=\sum_k \binom{n}{k}x^k$. **Engine 1 (substitute):** $f(1)=2^n$; $f(-1)=0$ (even/odd halves 2^{n-1} each); $f(2)=3^n$, i.e. $\sum\binom{n}{k}2^k=59049$ at $n=10$. **Engine 2 (differentiate/integrate):** the operator $x\frac{d}{dx}$ stamps k onto each term:

$$\sum k\binom{n}{k}=n2^{n-1}, \qquad \sum k(k-1)\binom{n}{k}=n(n-1)2^{n-2}$$

$$\sum k^2\binom{n}{k}=n(n+1)2^{n-2}$$

($k^2=k(k-1)+k$; at $n=8$: $8\cdot 9\cdot 2^6=4608$.) Divide-by-$(k+1)$ is the integral's fingerprint: $\sum\frac{\binom{n}{k}}{k+1}=\int_0^1(1+x)^n dx=\frac{2^{n+1}-1}{n+1}$, and $\sum\frac{(-1)^k}{k+1}\binom{n}{k}=\frac{1}{n+1}$.

INTUITION

Every identity is also a counting story

$\sum k\binom{n}{k}=n2^{n-1}$ counts committees with a chair: choose the chair (n), then any subset of the rest (2^{n-1}). Engines and stories are two proofs of one fact — quote whichever the question rewards.

KEY RESULT

Engine 3: products of coefficients are one coefficient

Vandermonde: $\sum_k\binom{r}{k}\binom{s}{m-k}=\binom{r+s}{m}$ — the x^m coefficient of $(1+x)^r(1+x)^s$, or committees from two groups. Specialisations: $\sum\binom{n}{k}^2=\binom{2n}{n}$ (at $n=10$: 184756; at $n=6$ via $\binom{6}{k}\binom{6}{6-k}$: $\binom{12}{6}=924$); $\sum\binom{n}{k}\binom{n}{k+1}=\binom{2n}{n+1}$; alternating squares $\sum(-1)^k\binom{n}{k}^2$ is the x^n coefficient of $(1-x^2)^n$: 0 for odd n, $(-1)^{n/2}\binom{n}{n/2}$ for even n. **Hockey-stick:** $\sum_{i=r}^{m}\binom{i}{r}=\binom{m+1}{r+1}$ (telescope Pascal), e.g. $\binom{4}{4}+\binom{5}{4}+\binom{6}{4}+\binom{7}{4}=\binom{8}{5}=56$.

COMMON TRAP

Common traps — identities

- Differentiating then evaluating at the wrong point, or dropping the outer x in $xf'(x)$ (a factor-x error away from $x=1$: $\sum k\binom{n}{k}2^k=2n\cdot 3^{n-1}$).
- Reaching for memorised identities when the engine derives them in one line — engines survive twists, lists do not.
- Vandermonde index mismatch: the second factor must be $\binom{s}{m-k}$ with the same k; check the target slot first.

LESSON 04 / 05

The ratio walk to the greatest term

KEY RESULT

Greatest term of $(a + bx)^n$ at a given x

For terms $t_k = \binom{n}{k} a^{n-k}(bx)^k$, walk the ratio $\frac{t_{k+1}}{t_k} = \frac{n-k}{k+1} \cdot \left|\frac{bx}{a}\right|$ until it crosses 1. Greatest term of $(2 + 3x)^8$ at $x = 1$: ratio $\frac{8-k}{k+1} \cdot \frac{3}{2} \geq 1$ iff $k \leq 4.4$, so terms rise through t_5 and fall after: $T_6 = \binom{8}{5} 2^3 3^5 = 108864$ (neighbours 90720, 81648 ✓). Use absolute values: "numerically greatest" means largest $|t_k|$.

KEY RESULT

Truncation: positivity turns expansions into inequalities

For $x \geq 0$ every term of $(1 + x)^n$ is non-negative, so any partial sum is a lower bound. Keeping $k \leq 1$: **Bernoulli** $(1 + x)^n \geq 1 + nx$; keeping $k \leq 2$ sharpens it by $\frac{n(n-1)}{2}x^2$. Bernoulli in fact holds for all $x > -1$ (induction proof — the truncation argument alone covers only $x \geq 0$ since odd terms flip sign).

The same term comparison traps $\left(1 + \frac{1}{n}\right)^n$: its k-th term is $\frac{1}{k!}\prod\left(1 - \frac{j}{n}\right) \leq \frac{1}{k!} \leq \frac{1}{2^{k-1}}$, giving $2 \leq \left(1 + \frac{1}{n}\right)^n \leq 3$ — e appears as the pinched limit.

KEY RESULT

Remainders and last digits: expand around the modulus

To compute A^n modulo m: write A as (multiple of m) $\pm$ small residue and expand — all terms carrying the multiple vanish mod m except the last few. Last digit of $7^{20} = 49^{10} = (50 - 1)^{10}$: every term except $(-1)^{10}$ carries a factor 50 (the $j = 1$ term is $-500 \equiv 0$ mod 100), so last digit 1. Similarly $6^{999} + 1 \equiv 1 + 1 = 2 \pmod 5$ since $6 \equiv 1 \pmod 5$.

KEY RESULT

Conjugate surds: integer parts without decimals

$(1 + \sqrt{2})^n + (1 - \sqrt{2})^n = 2\sum_j \binom{n}{2j} 2^j$ is an integer (odd $\sqrt{2}$ powers cancel by the binomial theorem itself), and $|1 - \sqrt{2}| < 1$ leaves only dust. For odd n the dust is negative, so the integer part of $(1 + \sqrt{2})^n$ is that integer: $(1 + \sqrt{2})^3$ has conjugate sum $2(1 + 6) = 14$, and integer part 14 (even, for every odd exponent). For even n the dust is positive and the integer part is the conjugate sum minus 1 (odd).

COMMON TRAP

Common traps — size questions

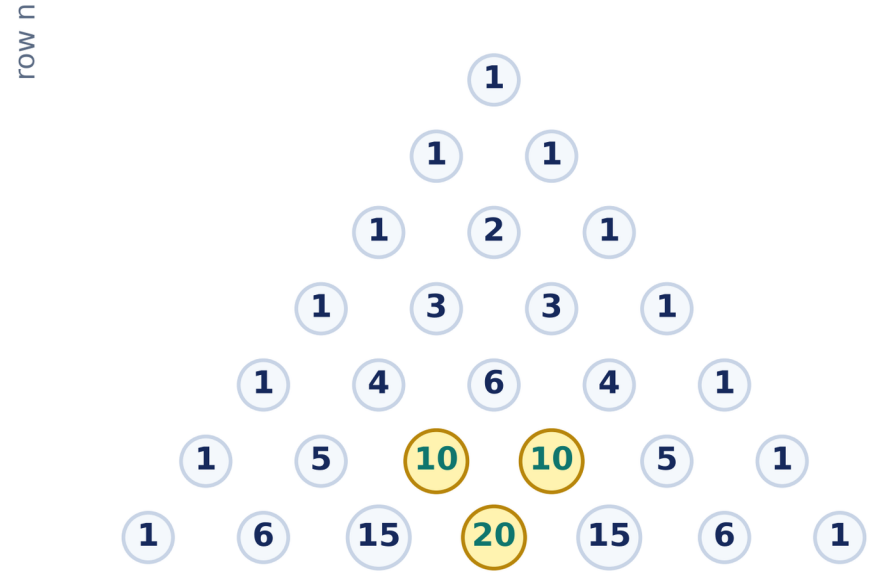
- Truncation claimed for negative x without the induction form of Bernoulli.
- Ratio walk without absolute values when terms alternate in sign.
- Modular truncation treated as approximation: it is exact in residue classes — but every dropped term must genuinely carry the modulus factor.

LESSON 05 / 05

Pascal's triangle is a recurrence, not just a table

The identity $\binom{n+1}{k} = \binom{n}{k-1} + \binom{n}{k}$ says that choosing k objects from $n+1$ either excludes a marked object or includes it. Each interior Pascal entry is therefore the sum of the two entries above it.

Two parents create each interior coefficient



READ THE FIGURE

The highlighted 10 + 10 = 20 shows C(6,3) = C(5,2) + C(5,3). Row n contains n + 1 entries indexed by k = 0 through n.

Row sums satisfy $\sum_{k=0}^{n} \binom{n}{k} = (1+1)^n = 2^n$. Alternating row sums vanish for $n \geq 1$ because $(1-1)^n = 0$. Both are substitutions in the same expansion, not unrelated identities.

Track every factor in the general term

For $(a+b)^n$, $T_{k+1} = \binom{n}{k} a^{n-k} b^k$. The term index is $k+1$, while the binomial lower index and the exponent of b are both k.

Term tracking for (2 + 3x)^4

k	C(4,k)	power of 2	power of 3x	complete term
0	1	2^4	(3x)^0	16
1	4	2^3	(3x)^1	96x
2	6	2^2	(3x)^2	216x^2
3	4	2^1	(3x)^3	216x^3
4	1	2^0	(3x)^4	81x^4

READ THE FIGURE

The coefficients include the numerical factors inside a and b. Binomial coefficients alone would give 1, 4, 6, 4, 1 and would not answer the coefficient question.

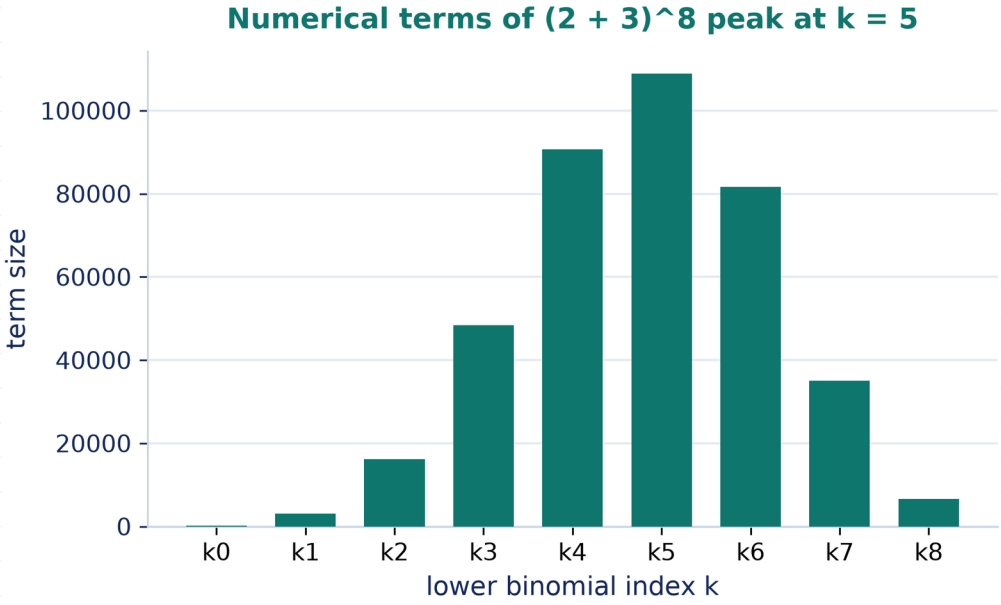
For a specified power x^r in $(p+qx^m)^n$, solve the exponent equation first and check that the resulting k is an integer between 0 and n. Only then calculate the coefficient.

Greatest numerical terms come from consecutive ratios

For $(2+3x)^8$ at $x=1$, write $t_k=\binom{8}{k}2^{8-k}3^k\ge 0$. Then

$$\frac{t_{k+1}}{t_k}=\frac{8-k}{k+1}\cdot\frac{3}{2}.$$

The terms increase while this ratio exceeds 1 and decrease after it falls below 1.



READ THE FIGURE

The largest value is t5 = 108864. It is the sixth written term because the written index is k + 1. The total sum is 5^8 = 390625.

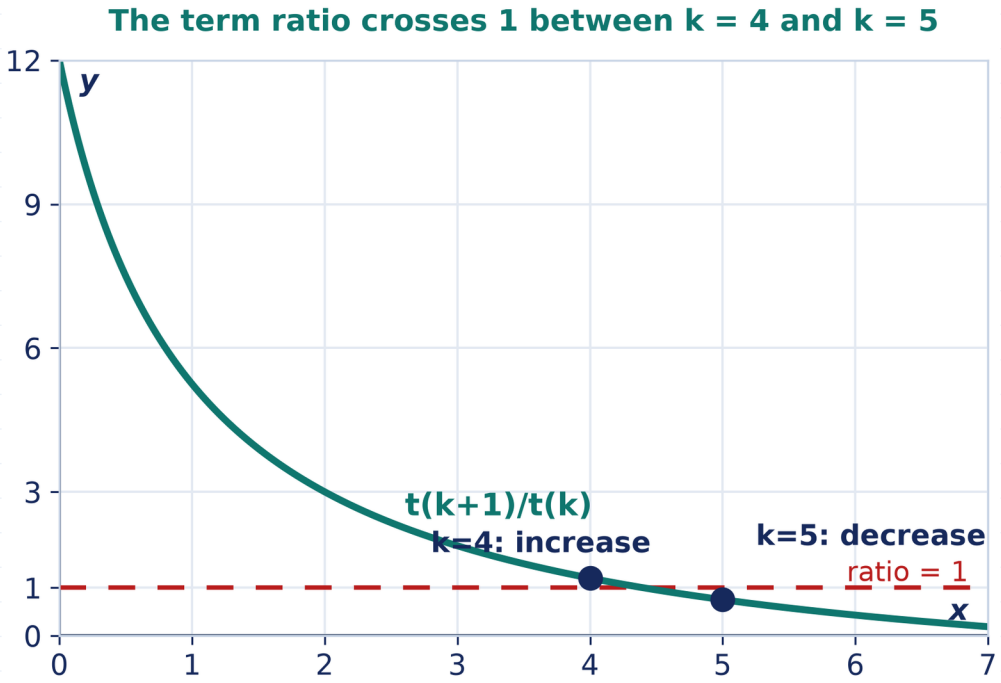
WORKED EXAMPLE

Check the peak directly

The ratio exceeds 1 exactly when $3(8-k)>2(k+1)$, or $k<22/5=4.4$. Thus $t_0<t_1<\cdots<t_5$, while $t_5>t_6>\cdots>t_8$. Direct evaluation gives $t_5=\binom{8}{5}2^3 3^5=108864$.

Read the ratio across one

A graph of the ratio makes the turnover visible. It is a guide for locating the maximum, not a replacement for checking integer indices and endpoint terms.



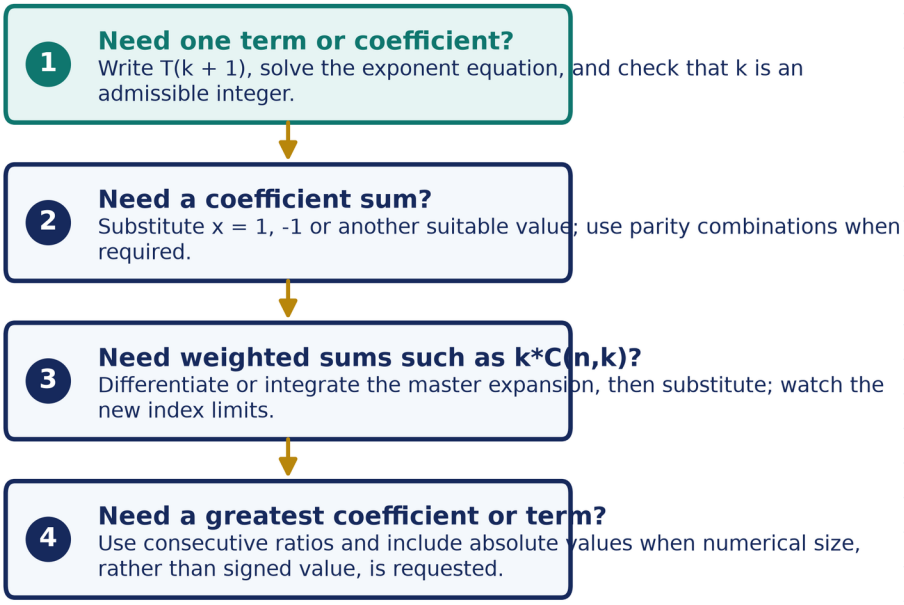
READ THE FIGURE

Only integer k values represent actual ratios. Since the continuous guide crosses 1 at k = 4.4, the last increase is from t4 to t5 and the peak is t5.

If terms can be negative, compare absolute values when "greatest numerical term" is requested. A greatest signed term and a term of greatest modulus are different questions.

Choose the right binomial operation

Match the operation to the requested quantity



READ THE FIGURE

Start from the same expansion and choose an operation matched to the question. This avoids memorising many disconnected identities.

KEY RESULT

Three final checks

Check the term index, include numerical factors from both binomial inputs, and state the admissible range of n, k, x. For truncation estimates, also state which side of the target the partial sum approaches and why.